Graphs & Transformations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

An ellipse has vertices at $(1, 6)$ and $(1, -4)$, and a focus at $(1, 4)$. Determine its standard equation and compute its eccentricity.

The vertices share $x = 1$, so the major axis is vertical. Center is the midpoint of the vertices: $(h, k) = \left(1, \frac{6+(-4)}{2}\right) = (1, 1)$. Semi-major axis: $a = 6 - 1 = 5$ (so $a^2 = 25$). Focal distance from center to $(1, 4)$: $c = 4 - 1 = 3$. For an ellipse, $b^2 = a^2 - c^2 = 25 - 9 = 16 \Rightarrow b = 4$. Standard equation: $\frac{(x-1)^2}{16} + \frac{(y-1)^2}{25} = 1$. Eccentricity: $e = \frac{c}{a} = \frac{3}{5} = 0.6$.

2 **[Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Eliminate the parameter to determine the Cartesian equation of the curve $x = 1 + 2\sec t$, $y = -2 + 3\tan t$. State the shape, center, and vertices.

Isolate the trigonometric terms: $\sec t = \frac{x-1}{2}$, $\tan t = \frac{y+2}{3}$. Apply the fundamental Pythagorean identity $\sec^2 t - \tan^2 t = 1$: $\left(\frac{x-1}{2}\right)^2 - \left(\frac{y+2}{3}\right)^2 = 1 \Rightarrow \frac{(x-1)^2}{4} - \frac{(y+2)^2}{9} = 1$. This is a horizontal hyperbola with:Center: $(h, k) = (1, -2)$Transverse semi-axis: $a = 2$Vertices: $(h \pm a, k) = (1 \pm 2, -2) = (-1, -2)$ and $(3, -2)$

3 **[Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

A radioactive substance follows the decay model $M(t) = M_0 e^{-kt}$. If 20% of the sample decays in 15 days, find the exact decay constant k and the substance's half-life in days to the nearest tenth.

If 20% decays in 15 days, then 80% remains at $t = 15$: $0.80M_0 = M_0 e^{-15k} \Rightarrow e^{-15k} = 0.80$. Take the natural logarithm of both sides: $-15k = \ln(0.80) \Rightarrow k = -\frac{\ln(0.80)}{15} = \frac{\ln(1.25)}{15} \approx 0.01488 \text{ day}^{-1}$. The half-life $t_{1/2}$ satisfies $e^{-kt_{1/2}} = 0.5$: $t_{1/2} = \frac{\ln 2}{k} = \frac{15\ln 2}{-\ln(0.80)} \approx \frac{15(0.6931)}{0.2231} \approx 46.6$ days.

4 **[Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

On an Argand diagram, the point P represents the complex number z such that $|z - (2 + 3i)| \leq 2$. Find the greatest possible value of $|z|$ for points in this region.

The locus $|z - (2 + 3i)| \leq 2$ represents a circular disc of radius $R = 2$ with center at $C(2, 3)$. The distance from the origin $O(0, 0)$ to the center C is: $|C| = \sqrt{2^2 + 3^2} = \sqrt{4 + 9} = \sqrt{13} \approx 3.6055$ The point on the disc farthest from the origin lies on the line passing through O and C, on the far boundary of the circle: $|z|_{\max} = |C| + R = \sqrt{13} + 2 \approx 3.606 + 2 = 5.61$ (3 s. f.)

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5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find all solutions to the trigonometric equation $2\sin^2 x - 3\sin x + 1 = 0$ on the domain $x \in [0, 2\pi)$.

Step 1: Factor the quadratic in $\sin x$: $(2\sin x - 1)(\sin x - 1) = 0$. Step 2: Solve each factor branch:Branch 1: $2\sin x - 1 = 0 \Rightarrow \sin x = \frac{1}{2}$. On $[0, 2\pi)$, $x = \frac{\pi}{6}$ (Quadrant I) or $x = \frac{5\pi}{6}$ (Quadrant II).Branch 2: $\sin x - 1 = 0 \Rightarrow \sin x = 1$. On $[0, 2\pi)$, $x = \frac{\pi}{2}$.The complete solution set is $x \in \left\{\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}\right\}$.

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Practice 1.2: An investor deposits 8,000 into a high-yield account earning an annual nominal interest rate of 4.8% compounded monthly. Determine the account balance after 6 years, and find the number of full years required for the investment to double in value.

1. $FV = PV\left(1 + \frac{r}{100k}\right)^{kn}$. Here $PV = 8000$, $r = 4.8$, $k = 12$, $n = 6$: $x10{,}660.80.
2. $For the investment to double$: 8000(1.004)^{12n} = 16000 \implies (1.004)^{12n} = 2
. $Take natural logs$: 12n \ln(1.004) = \ln(2) \implies n = \frac{\ln 2}{12 \ln(1.004)} \approx \frac{0.693147}{12(0.003992)} \approx \frac{0.693147}{0.0479} \approx 14.47 years. Therefore, 15 full years are required for the balance to double.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve the logarithmic equation $\log_2(x + 3) + \log_2(x - 3) = 4$ for all real solutions x. State any extraneous roots.

Step 1: State domain restrictions: $x + 3 > 0 \Rightarrow x > -3$ and $x - 3 > 0 \Rightarrow x > 3$, so $x > 3$. Step 2: Combine logarithms: $\log_2[(x + 3)(x - 3)] = 4 \Rightarrow \log_2(x^2 - 9) = 4$. $x^2 - 9 = 2^4 = 16 \Rightarrow x^2 = 25 \Rightarrow x = \pm 5$. Step 3: Eliminate extraneous solutions: $x = -5$ violates $x > 3$ (producing negative arguments), so it is extraneous. The unique valid solution is $x = 5$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the angle θ between the vectors $\mathbf{u} = \langle 1, \sqrt{3} \rangle$ and $\mathbf{v} = \langle 2, 0 \rangle$ in degrees.

Step 1: Compute dot product and magnitudes: $\mathbf{u} \cdot \mathbf{v} = (1)(2) + (\sqrt{3})(0) = 2$.
$||\mathbf{u}|| = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{1 + 3} = 2, \quad ||\mathbf{v}|| = \sqrt{2^2 + 0^2} = 2.$ Step 2: Solve for $\cos\theta$:
$\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{u}||||\mathbf{v}||} = \frac{2}{(2)(2)} = \frac{2}{4} = \frac{1}{2} \Rightarrow \theta = \arccos(1/2) = 60^\circ$.

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9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Verify the trigonometric identity: $\frac{\sin(2x)}{\sin x} - \frac{\cos(2x)}{\cos x} = \sec x$.

Expand the double-angle expressions on the left-hand side: $\text{LHS} = \frac{2\sin x\cos x}{\sin x} - \frac{2\cos^2 x - 1}{\cos x}$. Cancel $\sin x$ in the first term and combine over a common denominator of $\cos x$:

$= 2\cos x - \frac{2\cos^2 x - 1}{\cos x} = \frac{2\cos^2 x - (2\cos^2 x - 1)}{\cos x}$. $= \frac{2\cos^2 x - 2\cos^2 x + 1}{\cos x} = \frac{1}{\cos x} = \sec x = \text{RHS}$. ✓

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A student calculates $\int_1^3 \frac{4}{2x+1}\,dx$ by rounding $\ln 7 \approx 1.95$ and $\ln 3 \approx 1.10$ at the start, obtaining $2(1.95 - 1.10) = 1.70$. State the exact value of the integral and calculate the absolute error introduced by premature rounding.

Integrate exactly: $\int_1^3 \frac{4}{2x+1}\,dx = [2\ln(2x+1)]_1^3 = 2\ln 7 - 2\ln 3 = 2\ln\left(\frac{7}{3}\right) \approx 1.6946\ldots$ The exact value is $2\ln\left(\frac{7}{3}\right)$. The premature rounding gives 1.70, yielding an absolute error of: $|1.70 - 1.6946| \approx 0.0054$ On Cambridge Paper 3, this premature rounding costs the final accuracy mark (A1) because the 3 s.f. answer is 1.69, not 1.70.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In ΔABC, $a = 8$, $b = 11$, and $A = 35^\circ$. Determine the number of possible triangles and find the measure of angle B for all valid cases to the nearest tenth of a degree.

Calculate the altitude $h = b\sin A = 11\sin 35^\circ \approx 11(0.5736) = 6.31$. Since $htwopossibletriangles. BytheLawofSines$: \sin B = \frac{b\sin A}{a} = \frac{6.31}{8} \approx 0.7887.x. Second solution (obtuse): $B_2 = 180^\circ - 52.1^\circ = 127.9^\circ$. Check validity: $A + B_2 = 35^\circ + 127.9^\circ = 162.9^\circ 0$. Both $B_1 \approx 52.1^\circ$ and $B_2 \approx 127.9^\circ$ are valid.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find the vertex, focus, and directrix of the parabola given by the general equation $y^2 + 8x - 4y + 28 = 0$.

Complete the square on the y-terms: $y^2 - 4y = -8x - 28 \Rightarrow (y^2 - 4y + 4) = -8x - 28 + 4$.

$(y-2)^2 = -8x - 24 = -8(x+3)$. This matches $(y-k)^2 = 4p(x-h)$ with:Vertex: $(h,k) = (-3, 2)$ Focal parameter: $4p = -8 \Rightarrow p = -2$ (opens horizontally to the left)Focus: $(h+p, k) = (-3 + (-2), 2) = (-5, 2)$Directrix: $x = h - p = -3 - (-2) \Rightarrow x = -1$

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The equation $x^3 - x - 1 = 0$ has a root near $x = 1.3$. (a) Show that the iterative formula $x_{n+1} = (x_n + 1)^{1/3}$ converges by considering $g'(x)$. (b) Starting with $x_1 = 1.3$, calculate x_2 and x_3 to 4 decimal places.

(a) Let $g(x) = (x+1)^{1/3}$. Differentiate: $g'(x) = \frac{1}{3}(x+1)^{-2/3} = \frac{1}{3(x+1)^{2/3}}$ Near the root $x \approx 1.3$: $g'(1.3) = \frac{1}{3(2.3)^{2/3}} \approx \frac{1}{3(1.742)} \approx 0.191$ Since \(|g'(x)| $x_1 = 1.3$ $x_2 = (1.3+1)^{1/3} = (2.3)^{1/3} \approx 1.3200$ $x_3 = (1.3200+1)^{1/3} = (2.3200)^{1/3} \approx 1.3238$

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14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Variables x and y satisfy $y = ax^b$. A graph of $\ln y$ against $\ln x$ is a straight line passing through $(0, 1.20)$ and $(4, 6.80)$. Find the values of the constants a and b to 3 significant figures.

Take natural logarithms of $y = ax^b$: $\ln y = \ln a + b\ln x$ This is a linear equation $Y = mX + c$ with $Y = \ln y$, $X = \ln x$, gradient $m = b$, and intercept $c = \ln a$. Gradient: $b = \frac{6.80 - 1.20}{4 - 0} = \frac{5.60}{4} = 1.40$ Vertical intercept gives $\ln a = 1.20$: $a = e^{1.20} \approx 3.32$ Therefore, $a = 3.32$ and $b = 1.40$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 17.01 Discrete Random Variables**

Find the 15th term and the sum of the first 20 terms of the arithmetic progression $4, 9, 14, 19, \ldots$.

Step 1: Identify progression parameters: First term $a_1 = 4$, common difference $d = 9 - 4 = 5$. Step 2: Compute a_{15}: $a_{15} = a_1 + (15 - 1)d = 4 + 14(5) = 4 + 70 = 74$. Step 3: Compute finite sum S_{20}:

$S_{20} = \frac{20}{2}[2a_1 + (20 - 1)d] = 10[2(4) + 19(5)] = 10[8 + 95] = 10(103) = 1030.$

16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 18.01 Binomial Distribution**

A radioactive isotope has a decay constant $k = 0.035\ \text{year}^{-1}$. (a) Find its half-life to the nearest tenth of a year. (b) If an initial sample contains $250\ \text{mg}$, how much remains after $40\ \text{years}$?

(a) Half-life calculation: $t_{1/2} = \frac{\ln 2}{k} = \frac{0.693147}{0.035} \approx 19.8\ \text{years}$. (b) Remaining mass at $t = 40$:

$A(t) = A_0 e^{-kt} = 250e^{-0.035(40)} = 250e^{-1.4} \approx 250(0.246597) \approx 61.65\ \text{mg}.$

17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 19.01 Normal Distribution**

A manufacturer claims that the mean breaking strength of cables is at least 80 kN. A sample of 16 cables gives $\bar{x} = 77.5$ kN with population SD 5 kN. Test this claim at 1% significance.

$H_0 : \mu = 80$, $H_1 : \mu < 80$. $Z = \frac{77.5 - 80}{5/4} = -\frac{2.5}{1.25} = -2$. Critical value at 1%: -2.326. Since $-2 > -2.326$, do not reject H_0. Insufficient evidence at 1% that mean breaking strength is below 80 kN.

18 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 2.01 Simultaneous Equations & Inequalities**

A sample of size 100 from a population (known $\sigma = 20$) gives $\bar{x} = 142$. Find a 99% CI for μ and state what "99% confidence" means.

$142 \pm 2.576 \cdot \frac{20}{\sqrt{100}} = 142 \pm 5.15 = (136.85,\ 147.15)$. "99% confidence" means: if this process were repeated many times, 99% of the intervals produced would contain the true μ.

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19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Coordinate Geometry

Lines ℓ_1 and ℓ_2 have direction vectors $\mathbf{d_1} = (2; 1; -1)$ and $\mathbf{d_2} = (1; -1; 3)$. Find the acute angle between them.

$\mathbf{d_1} \cdot \mathbf{d_2} = 2 - 1 - 3 = -2$. $|d_1| = 6$, $|\mathbf{d_2}| = \sqrt{11}$. $\cos\theta = \frac{2}{\sqrt{66}} \Rightarrow \theta = \arccos\left(\frac{2}{\sqrt{66}}\right) \approx 75.7°$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Trigonometry

An IB AA SL student scores 64/80 on Paper 1 (40%), 72/80 on Paper 2 (40%), and 16/20 on the Internal Assessment (20%). Calculate the student's overall weighted percentage.

Paper 1: $\frac{64}{80} = 80\%$. Paper 2: $\frac{72}{80} = 90\%$. IA: $\frac{16}{20} = 80\%$. Overall weighted percentage: $0.40(80) + 0.40(90) + 0.20(80) = 32 + 36 + 16 = 84\%$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Describe the transformation that maps the curve $y = f(x)$ onto $y = f(x - 3) + 5$.

a. **Translation by vector** $[3, 5]$ ✓

b. Translation by vector $[-3, 5]$

c. Translation by vector $[3, -5]$

d. Translation by vector $[-3, -5]$

Replacing x by $x - 3$ translates 3 units right ($+x$). Adding 5 translates 5 units up ($+y$). Vector is $[3, 5]$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The curve $y = x^2$ is translated by vector $[-2, -4]$. What is the equation of the resulting curve?

a. $y = (x + 2)^2 - 4$ ✓

b. $y = (x - 2)^2 - 4$

c. $y = (x + 2)^2 + 4$

d. $y = x^2 - 2x - 4$

A shift of -2 horizontally replaces x with $x + 2$, and -4 vertically subtracts 4: $y = (x + 2)^2 - 4$.

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the coordinates of the turning point of $y = 4 - (x - 1)^2$.

a. $(1, 4)$ ✓

b. $(-1, 4)$

c. $(1, -4)$

d. $(4, 1)$

The vertex occurs when $x - 1 = 0 \Rightarrow x = 1$, giving $y = 4$. Point is $(1, 4)$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Describe the transformation that maps $y = f(x)$ onto $y = -f(2x)$.

a. **Reflection in x-axis followed by horizontal stretch scale factor $1/2$** ✓

b. Reflection in y-axis followed by vertical stretch scale factor 2

c. Reflection in x-axis followed by horizontal stretch scale factor 2

d. Reflection in y-axis followed by horizontal stretch scale factor $1/2$

$-f(\cdot)$ is a reflection in the x-axis. $f(2x)$ is a horizontal stretch by scale factor $1/2$ parallel to the x-axis.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The point $(4, 12)$ lies on the curve $y = f(x)$. What are the coordinates of the corresponding point on $y = 3f(x/2)$?

a. $(8, 36)$ ✓

b. $(2, 36)$

c. $(8, 4)$

d. $(2, 4)$

x-coordinate is multiplied by 2: $4 \times 2 = 8$. y-coordinate is multiplied by 3: $12 \times 3 = 36$. Point is $(8, 36)$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which equation represents the reflection of the curve $y = 2^x$ in the y-axis?

a. $y = 2^{-x}$ ✓

b. $y = -2^x$

c. $y = -2^{-x}$

d. $y = \log_2(x)$

A reflection in the y-axis replaces x by $-x$, yielding $y = 2^{-x} = (1/2)^x$.

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The graph of $y = x^3$ undergoes a stretch scale factor 2 parallel to the y-axis, followed by a translation by vector $[1, -3]$. Find its equation.

a. $y = 2(x-1)^3 - 3$ ✓

b. $y = (2x-1)^3 - 3$

c. $y = 2(x+1)^3 - 3$

d. $y = 2x^3 - 5$

Stretch by 2 parallel to y-axis: $y = 2x^3$. Then translation $[1, -3]$: $y = 2(x-1)^3 - 3$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A curve with equation $y = g(x)$ has a local minimum at $(3, -5)$. Where is the corresponding local extremum on the curve $y = -2g(x+1)$?

a. **Local maximum at** $(2, 10)$ ✓

b. Local minimum at $(4, 10)$

c. Local maximum at $(4, -10)$

d. Local minimum at $(2, -10)$

Horizontal shift: $x = 3 - 1 = 2$. Vertical stretch and reflection: $y = -2(-5) = 10$. Reflection flips minimum into a maximum at $(2, 10)$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the asymptotes of the rational function $y = \frac{2x+1}{x-3}$.

a. $x = 3$ **and** $y = 2$ ✓

b. $x = -3$ and $y = 2$

c. $x = 3$ and $y = 1$

d. $x = 1/2$ and $y = 3$

Vertical asymptote when denominator is zero: $x = 3$. Horizontal asymptote as $x \to \pm\infty$: $y = 2/1 = 2$.

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the coordinates of all intersection points of $y = (x-2)^2(x+1)$ with the coordinate axes.

a. **x-intercepts at $(2, 0)$ and $(-1, 0)$; y-intercept at $(0, 4)$** ✓

b. x-intercepts at $(-2, 0)$ and $(1, 0)$; y-intercept at $(0, -4)$

c. x-intercept at $(2, 0)$; y-intercept at $(0, 4)$

d. x-intercepts at $(2, 0)$ and $(-1, 0)$; y-intercept at $(0, -2)$

For $y = 0$: $x = 2$ (double root) and $x = -1$. For $x = 0$: $y = (-2)^2(1) = 4$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The curve $y = \frac{a}{x+b}$ passes through $(1, 6)$ and $(4, 2)$. Find the values of a and b.

a. $a = 12, b = 1$ ✓

b. $a = 6, b = 0$

c. $a = 18, b = 2$

d. $a = 8, b = 0$

$6 = a/(1+b) \Rightarrow a = 6 + 6b$. $2 = a/(4+b) \Rightarrow a = 8 + 2b$. Thus $6 + 6b = 8 + 2b \Rightarrow 4b = 2 \Rightarrow b = 1/2$ and $a = 9$. Wait: if $a = 12, b = 1$: $12/2 = 6, 12/5 = 2.4 \neq 2$. Let's solve:

$a = 6(1+b) = 2(4+b) \Rightarrow 6 + 6b = 8 + 2b \Rightarrow 4b = 2 \Rightarrow b = 1/2, a = 9$. Adjusted choices: $a = 9, b = 0.5$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Determine the number of real solutions to the equation $|x^2 - 4| = 3$.

a. **4 solutions** ✓

b. 2 solutions

c. 3 solutions

d. 0 solutions

$x^2 - 4 = 3 \Rightarrow x^2 = 7 \Rightarrow x = \pm\sqrt{7}$ (2 solutions). $x^2 - 4 = -3 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$ (2 solutions). Total 4 distinct real solutions.

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the function $f(x) = x^2 - 4x + 3$ and the function $g(x) = x - 1$. The graphs of $y = f(x)$ and $y = g(x)$ intersect at points A and B. One of these intersection points is $(1, 0)$. Find the coordinates of the other intersection point.

a. **(4, 3) ✓**

b. (3, 2)

c. (5, 4)

d. (2, 1)

To find the intersection points of the two graphs, we set $f(x) = g(x)$. $x^2 - 4x + 3 = x - 1$ Subtract x from both sides: $x^2 - 5x + 3 = -1$ Add 1 to both sides: $x^2 - 5x + 4 = 0$ Factor the quadratic equation: $(x - 1)(x - 4) = 0$ This gives us two possible x-coordinates for the intersection points: $x = 1$ and $x = 4$. We are given that one intersection point has an x-coordinate of 1. To find the corresponding y-coordinate, we can substitute $x = 1$ into either $f(x)$ or $g(x)$. Using $g(x) = x - 1$, we get.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of $g(x) = f(x + 1) - 5$ is obtained from $f(x)$ by:

a. Shifting 1 units left and 5 units down

b. **Shifting 1 units right and 5 units down ✓**

c. Shifting 1 units right and 5 units up

d. Reflecting over the x-axis

Replacing x with $x - 1$ shifts horizontally by 1 (right if positive); adding -5 shifts vertically by -5.

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the functions $f(x) = |x - a|$ and $g(x) = b$, where a and b are constants. The graph of $y = f(x)$ is a V-shaped graph with its vertex at $(a, 0)$. The graph of $y = g(x)$ is a horizontal line. If the graphs of $y = f(x)$ and $y = g(x)$ intersect at exactly one point, what is the relationship between a and b for the graphs to intersect at a single point?

a. a = b

b. a = -b

c. **b = 0 ✓**

d. a = 0

The function $f(x) = |x - a|$ represents an absolute value function whose graph is a V-shape with its vertex at the point $(a, 0)$. The function $g(x) = b$ represents a horizontal line at the height $y = b$. For the graphs of $y = f(x)$ and $y = g(x)$ to intersect at exactly one point, the horizontal line $y = b$ must pass through the vertex of the V-shaped graph $y = |x - a|$. The vertex of $y = |x - a|$ is at $(a, 0)$. Therefore, for the intersection to occur at a single point, the y-coordinate of the horizontal line must be equal.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Convert 210° to radians, giving your answer as an exact multiple of π.

a. $\frac{5\pi}{3}$

b. $\frac{\pi}{2}$

c. $\frac{7\pi}{6}$ ✓

d. $\frac{7\pi}{4}$

Working through this trigonometry step: $210^\circ \times \frac{\pi}{180} = \frac{7\pi}{6}$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For what value of k is $f(x) = -2x + k$ (for $x < 5$) and $f(x) = x^2$ (for $x \geq 5$) continuous at $x = 5$?

a. 36

b. 34

c. -35

d. **35 ✓**

Continuity requires $-2x + k$ at $x = 5$ to equal $(5)^2$: $-10 + k = 25$, so $k = 35$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

A root of $f(x)$ lies in $[-6, -3]$, where $f(-6)$ is negative and $f(-3)$ is positive. After one bisection step (midpoint $m = \frac{-6-3}{2} = -\frac{9}{2}$), $f(m)$ is negative. Which sub-interval contains the root?

a. $[-6, -\frac{9}{2}]$

b. $[-\frac{9}{2}, -3]$ ✓

c. [-6,-3]

d. [-7,-6]

Since $f(m)$ is negative, the sign change lies in $[-\frac{9}{2}, -3]$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Coordinate Geometry

Given $f(x) = (x-2)^2(x+1)$, does the graph touch or cross the x-axis at $x = 2$?

a. The graph does not intersect the x-axis there

b. Not enough information

c. **Touches (does not cross)** ✓

d. Crosses

Multiplicity of $x = 2$ is 2 (even), so the graph touches the x-axis.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Trigonometry

If $f(x) = \sqrt{x-5}$ and $g(x) = x^2 - 4$, what is the domain of $f(g(x))$?

a. $-3 \le x \le 3$

b. $x \le -3$ **or** $x \ge 3$ ✓

c. $x \ge 3$

d. All real numbers

Need $x^2 \ge 9$, so $|x| \ge 3$, giving $x \le -3$ or $x \ge 3$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Differentiation & Integration

A function is defined by $f(x) = x^2$ for $x < 3$ and $f(x) = -2x + 1$ for $x \ge 3$. Find $f(3)$.

a. -6

b. -4

c. 9

d. **-5** ✓

Since $x = 3 \ge 3$, use $f(x) = -2x + 1$: $f(3) = -2(3) + 1 = -5$.

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 7.01 Statistics & Probability

Determine whether $f(x) = x^5 + 2x$ is even, odd, or neither.

a. Even

b. **Odd ✓**

c. Neither

d. Cannot be determined

Every term has odd degree, so $f(-x) = -f(x)$: odd.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 8.01 Functions: Composite & Inverse

In the diagram, triangle ABC is shown with coordinates A(2, 1), B(7, 1), and C(5, 5). What is the length of the side AC?

a. 4

b. 5

c. $2\sqrt{5}$ ✓

d. $\sqrt{20}$

To find the length of the side AC, we use the distance formula between points A(1, 2) and C(3, 6). The distance formula is given by $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. Substituting the coordinates of A and C: $AC = \sqrt{(3-1)^2 + (6-2)^2}$ $AC = \sqrt{(2)^2 + (4)^2}$ $AC = \sqrt{4 + 16}$ $AC = \sqrt{20}$ To simplify $\sqrt{20}$, we find the largest perfect square factor of 20, which is 4. $\sqrt{20} = \sqrt{4 \times 5} = \sqrt{4} \times \sqrt{5} = 2\sqrt{5}$. Therefore, the length of side AC is.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Circular Measure

The diagram shows part of the curve $y = 9 - x^2$ for $x \geq 0$ and the tangent to the curve at the point $P(2, 5)$. Find the exact area of the finite region bounded by the curve, the tangent at P, and the x-axis.

a. $\frac{11}{24}$ ✓

b. $\frac{125}{192}$

c. $\frac{8}{3}$

d. $\frac{25}{8}$

1. **Find the equation of the tangent at $P(2, 5)$:** Given $y = 9 - x^2$, differentiating with respect to x gives: $\frac{dy}{dx} = -2x$ At $x = 2$, the gradient of the tangent is $m = -2(2) = -4$. The equation of the tangent line is: $y - 5 = -4(x - 2) \Rightarrow y = 13 - 4x$ 2. **Determine the x-intercepts:** - Tangent: Set $y = 0 \Rightarrow 13 - 4x = 0 \Rightarrow x = \frac{13}{4} = 3.25$. - Curve: Set $y = 0 \Rightarrow 9 - x^2 = 0 \Rightarrow x = 3$ (since $x \geq 0$). 3. **Calculate the area of the bou.

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Indices, Surds & Quadratics

The diagram shows part of the curve $y = ax^3 + bx^2 + 9x + 1$, where a and b are constants. The curve has a stationary point at $(1, 5)$. Find the coordinates of the other stationary point on the curve.

a. $(3, 1)$ ✓

b. $(3, 0)$

c. $(3, -1)$

d. $(-3, 1)$

To find the coordinates of the other stationary point, we first determine the unknown parameters a and b. **Step 1: Use the given coordinates of the point on the curve** Since the point $(1, 5)$ lies on the curve: $5 = a(1)^3 + b(1)^2 + 9(1) + 1\ 5 = a + b + 10 \Rightarrow a + b = -5 \quad - - -$ (Equation 1) **Step 2: Differentiate to find $\frac{dy}{dx}$** $\frac{dy}{dx} = 3ax^2 + 2bx + 9$ **Step 3: Apply the stationary condition at $x = 1$** At a stationary poi.

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